

Assignment-1

Basic Variable Swapping

1. Switch values of two integers

Input:

```
n1 = 20,
```

```
n2 = 30
```

Output:

```
n1 = 30
```

```
n2 = 20
```

2. Switch values of two strings (characters)

Input:

```
char1 = "hello"
```

```
char2 = "java"
```

Output:

```
char1 = "java"
```

```
char2 = "hello"
```

3. Switch one string value with one integer value

Input:

```
n1 = 200,
```

```
char2 = "java"
```

Output:

```
n1 = "java"
```

```
char2 = 200
```

Banking and Transactions

5. Update balance after deposit

Initial balance: `current_balance = 10000`

Deposit amount: `deposit_balance = 5000`

Output:

- Before deposit: `current_balance = 10000`
- After deposit: `current_balance = 15000`

6. Update balance after withdrawal

Before: `balance = 12000`

Withdrawal: `3000`

After: ?

7. Increase shopping cart items by 3

Before: `cart_items = 5`

After: ?

8. Apply a 20% discount to a price

Before: `price = 1000`

After: ?

Academic Calculations

9. Calculate student percentage

Input: `obtain_marks = 430, out_of = 500`

Output: `Percentage = ?`

10. Calculate total and average of 4 subjects

Input Example:

- Subject 1: 80
- Subject 2: 75
- Subject 3: 90
- Subject 4: 85

11. Calculate average of 3 numbers

Input: `num1 = 25, num2 = 35, num3 = 45`

Output: `Average = ?`

Finance & Business Calculations

12. Calculate profit or loss percentage

Input: `cost_price = 500, selling_price = 600`

Output: `Profit or Loss = ?`

13. Calculate simple interest

Input: `principal = 10000, rate = 5, time = 2`

Output: `Simple Interest = ?`

14. Calculate compound interest

Input: `principal = 10000, rate = 5, time = 2`

Output: `Compound Interest = ?`

15. Calculate tax on income

Input: `income = 500000, tax_rate = 10`

Output: `Tax = ?`

16. Calculate percentage increase or decrease

Input: `initial_value = 200, final_value = 250`

Output: `Percentage Change = ?`

Conversions

17. Convert boolean to integer

Input: `True`

Output: `?`

18. Convert float to string

Input: `45.67`

Output: `?`

19. Convert 20°C to Fahrenheit

Input: `20°C`

Output: `?`

20. Convert 50°F to Celsius

Input: `50°F`

Output: `?`

21. Convert integer to binary

Input: `25`

Output: `?`

Geometry

21. **Calculate area of a triangle**

Input: `base = 10, height = 6`

Output: `Area = ?`

22. **Calculate perimeter of a square**

Input: `side = 9`

Output: `Perimeter = ?`

23. Calculate diameter of a circle

Input: radius = 14

Output: Diameter = ?

24. Calculate volume of a cube

Input: side = 5

Output: Volume = ?

25. Calculate surface area of a cuboid

Input: l = 4, b = 3, h = 2

Output: Surface Area = ?

Mathematical Expressions

26. Square of sum: $(x + y)^2$

Input: x = 5, y = 7

Output: ?

27. Simplify expression: $x^2 - 4x + 4$

Input: x = 3

Output: ?

25. Evaluate: $(a + b)(a - b)$

Input: a = 6, b = 2

Output: ?

29. Sum of cubes: $a^3 + b^3$

Input: a = 1, b = 2

Output: ?

30. Simplify: $(x - y)^2$

Input: x = 10, y = 6

Output: ?

31. Difference of cubes: $x^3 - y^3$

Input: x = 4, y = 1

Output: ?

String Operations

32.If user input is:

Name: Dev

Age: 25

City: Jaipur

Hobby: Cooking

Then Output is :

Meet Dev, a 25-year-old enthusiast from Jaipur.

When not busy with daily tasks, Dev loves spending time cooking.

Life in Jaipur keeps Dev energetic and curious every single day.

With coding as a passion, the future looks creative and inspiring for Dev in the Jaipur City.

33.Create a Python program that asks the user for the following:

- Full Name
- Profession
- Favorite Quote
- Years of Experience

Output format:

Name : <Full Name>

Profession : <Profession>

Experience : <Years of Experience> years

Quote : "<Favorite Quote>"

34. Ask the user:

- Movie Name
- Viewer Name
- Seat Number
- Show Time

Output format:

 Movie Ticket 

Movie : <Movie Name>

Name : <Viewer Name>

Seat No : <Seat Number>

Time : <Show Time>

Enjoy the show!
